Software Requirements Specification

for

Mess Automation System

Version <1.0>

Prepared by

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
Initial draft(v1.0)	B Mahath Priyanshi Meena Shreyansh Basavakiran Rishith Rajesh Parth Jhanani Rohan Siddhant	Initiated the Document and added all necessary details	23/01/2026

1 Introduction

1.1 Product Scope

The Mess Automation System is a web-based software solution designed to streamline and digitize the operations of a hostel mess. The primary objective of this system is to eliminate

inefficiencies associated with traditional mess management practices such as long queues, manual entry of students, and error-prone record keeping. By automating daily mess activities, the system aims to provide a faster, more organized, and user-friendly experience for both students and mess administrators.

This system enables students to authenticate themselves digitally, participate in menu selection through a voting mechanism, and manage their mess-related activities via a dedicated student dashboard. It also integrates billing and extra purchase management, ensuring accurate and transparent financial records. For mess managers, the system provides an admin dashboard to monitor student participation, manage menus, track billing, and collect feedback efficiently. Overall, the Mess Automation System improves operational efficiency, reduces manual workload, minimizes waiting time, and enhances transparency and satisfaction within the hostel mess ecosystem.

The main objectives of the Mess Automation System are to:

- Reduce long queues by automating student entry and mess-related processes.
- Eliminate manual record keeping through digital authentication and billing.
- Enable students to vote for mess menus and provide structured feedback.
- Integrate mess billing and extra purchase tracking in a centralized system.
- Provide separate dashboards for students and mess managers for better control and monitoring.

- Improve overall efficiency, transparency, and user experience in hostel mess operations.

1.2 Intended Audience and Document Overview

1.2.1 Intended Audience

This Software Requirements Specification (SRS) document is intended for multiple stakeholders involved in the design, development, testing, and evaluation of the Mess Automation System. The

primary audience of this document includes developers, testers, and instructors associated with the project.

Developers will use this document as a reference to understand the system's functional and non-functional requirements, system behavior, and constraints during implementation. Testers can refer to this SRS to design test cases, validate system functionality, and ensure that the system meets the specified requirements. Instructors and evaluators can use this document to review the project scope, assess requirement completeness, and understand the overall system design and objectives.

This document serves as a central reference throughout the project lifecycle, ensuring that all stakeholders share a common understanding of the system requirements.

1.2.2 Document Organization and Overview

This document is organized into multiple sections, each focusing on a specific aspect of the Mess Automation System. The structure of the document is designed to provide a clear and systematic understanding of the system requirements.

- **Section 2: Overall Description**
  This section provides a high-level overview of the system, including its purpose, operating environment, user characteristics, assumptions, and constraints. It helps readers understand the context in which the system will operate.
- **Section 3: Specific Requirements**
  This section describes the detailed functional requirements of the system. It outlines system features, user interactions, and software interfaces required for the implementation of the Mess Automation System.
- **Section 4: Non-Functional Requirements**
  This section specifies the non-functional aspects of the system such as performance requirements, security considerations, usability, and reliability constraints.

Readers are encouraged to begin with this introduction section to gain an overall understanding of the system, and then proceed to the sections most relevant to their role.

1.3 Definitions, Acronyms and Abbreviations

Acronym / Abbreviation	Full Form	Description
API	Application Programming Interface	Enables communication between frontend and backend components.
B/L/D	Breakfast / Lunch / Dinner	Meal types supported by the mess system.
DBMS	Database Management System	Software used to store and manage system data securely.
FR	Functional Requirement	A requirement that specifies system functionality.
NFR	Non-Functional Requirement	A requirement that specifies quality attributes such as performance and security.
RBAC	Role-Based Access Control	Access control mechanism based on user roles.
SRS	Software Requirements Specification	A document that describes the functional and non-functional requirements of the system.
UI	User Interface	The visual interface through which users interact with the system.

1.4 Document Conventions.

This document follows the IEEE Software Requirements Specification (SRS) formatting standards. It uses a standard font with size 11 or 12, single spacing, and 1-inch margins on all sides. Headings and subheadings adhere to the IEEE template conventions, with italics used for emphasis where necessary to ensure clarity and consistency throughout the document.

1.5 References and Acknowledgments

1. Digital Tools and Resources:

- **Figma:** Used for designing the user interface,flowchart and creating interactive prototypes. https://www.figma.com
- **Moqups:** Employed to our use case models. https://moqups.com
- **LibreOffice (CC):** Utilized for drafting and formatting the document within the Computer Centre environment.
- **Course Content:** We referred to the instructional videos and lecture slides provided by our professor to ensure the document meets the required academic standards.

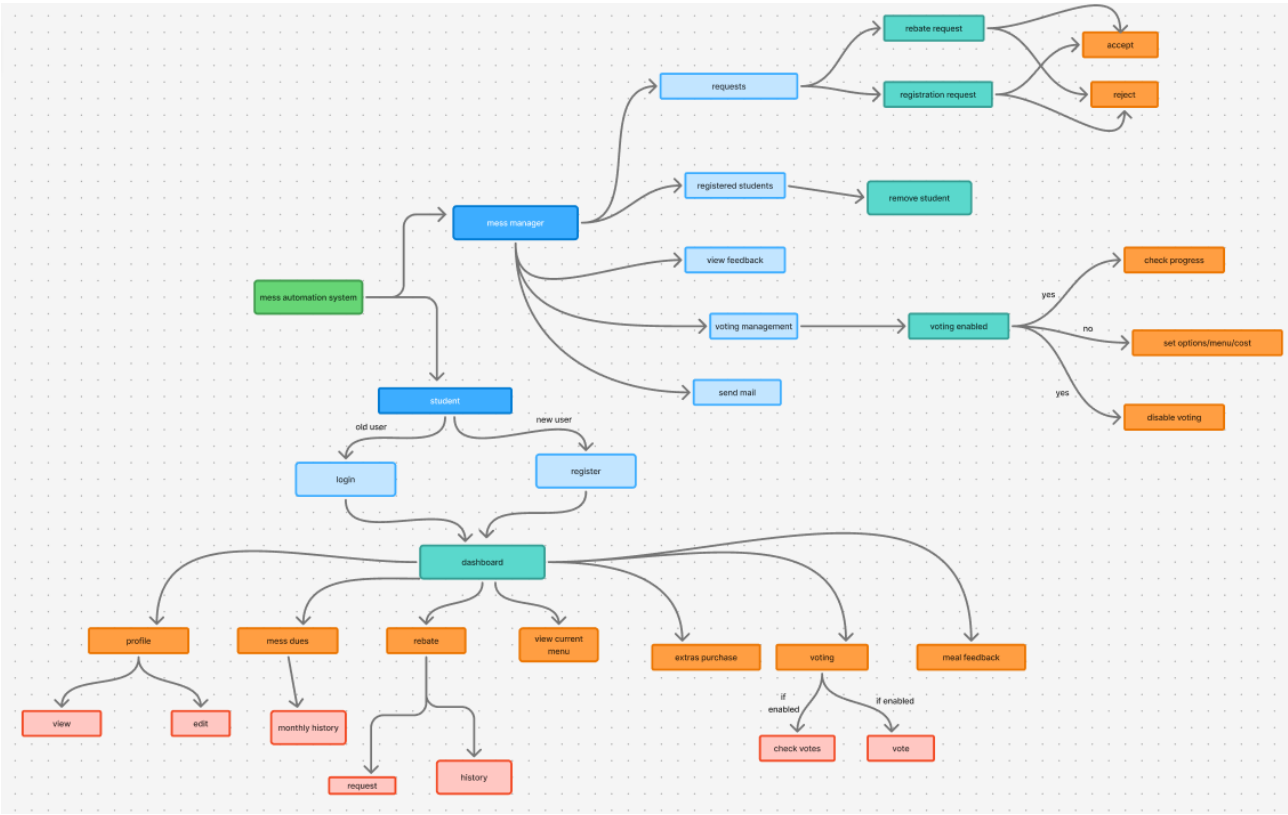
- **Google Search:** Used to research existing mess management workflows and common practices in automation systems.

2. Acknowledgments:

- We would like to thank our instructor, **Prof. Indranil Saha**, for providing the project template and for his detailed lecture videos that helped us understand how to structure this document.
- We also appreciate the support of our TA, **Ujjwal Kajal**, for discussing the project with us and providing helpful suggestions to improve the system's requirements.

2 Overall Description

2.1 Product Overview



The Mess Automation System is a web-based platform developed to automate and manage the routine operations of a hostel mess. The system aims to reduce manual effort, minimize billing errors, and improve transparency by replacing traditional paper-based and manual processes with a centralized digital solution. It provides secure and role-based access for students and mess managers and is designed to be scalable across multiple messes within an institution.

From a student perspective, the system allows users to log in using institutional credentials with optional face-recognition–based authentication. Students can view mess dues, purchase extra food items, track monthly expenses, participate in menu voting, request rebates, and submit feedback through a personalized dashboard. Automated email summaries further enhance transparency by keeping students informed about their monthly extra food consumption.

From an administrative perspective, mess managers can manage student registrations, handle rebate requests, configure menus and pricing, control voting processes, and review feedback. A

centralized database ensures accurate recording of all transactions, votes, and feedback. The figure below presents a high-level overview of the system architecture and illustrates the interaction between students, mess managers, and the core functional components of the Mess Automation System.

2.2 Product Functionality

I. User Authentication and Access Control

A. Secure login using institutional credentials

B. Optional face-recognition–based authentication for faster access

C. Role-based access for students and mess managers

D. Password recovery and session management

II. Student Module

A. Dashboard

View mess dues and monthly transaction summary

Access current mess menu and announcements

B. Extras Food Management

Purchase extra food items

View extras purchase history and pending dues

C. Menu Voting System

View available dish options

Vote for preferred dishes during the voting window

View current voting status

D. Rebate Management

Submit rebate requests

View rebate request history and status

E. Profile Management

View personal details

Update profile information and change password

F. Feedback System

Submit feedback on food quality and mess services

III. Mess Manager Module

A. Student Management

Approve or reject student registration requests

Remove registered students

B. Inventory and Menu Management

Add or remove dishes

Update raw material availability and pricing

C. Voting Management

Enable or disable voting windows

Monitor voting progress

Finalize menu based on voting results

D. Request Handling

Approve or reject rebate requests

E. Communication and Reportin

Send notifications or emails to students

View submitted feedback

Generate automated monthly reports

IV. Data Management and Analytics

Centralized database for storing transactions, votes, and feedback

Basic analytics for consumption tracking and decision support

2.3 Design and Implementation Constraints

The development and deployment of the Mess Automation System are subject to the following constraints:

Hardware Constraints

The system requires computer or tablet or mobile at the mess entry point for student authentication and verification of their coupons.
Reliable Camera Device must be available for scanning student face ID.

Software Constraints

The system must be developed as a web-based application compatible with modern browsers.
The system must operate within the campus intranet for security, with optional secure external access for administrators.

Security Constraints

All user data, especially billing and personal information, must be encrypted in transit and at rest.
Access to admin features must be restricted via role-based authentication.

2.4 Assumptions and Dependencies

The successful implementation and operation of the Mess Automation System rely on the following assumptions and dependencies:

2.4.1 Assumptions

Students will possess a valid student ID and credentials for authentication, and a mobile phone.
The hostel mess operates on a fixed schedule (breakfast, lunch, dinner) that can be configured in the system.
Network connectivity will be available at mess entry points and admin offices.

2.4.2 Dependencies

Hardware Dependencies

Computers or tablets or mobile phones at mess counters, with a reliable camera device.

External Dependencies

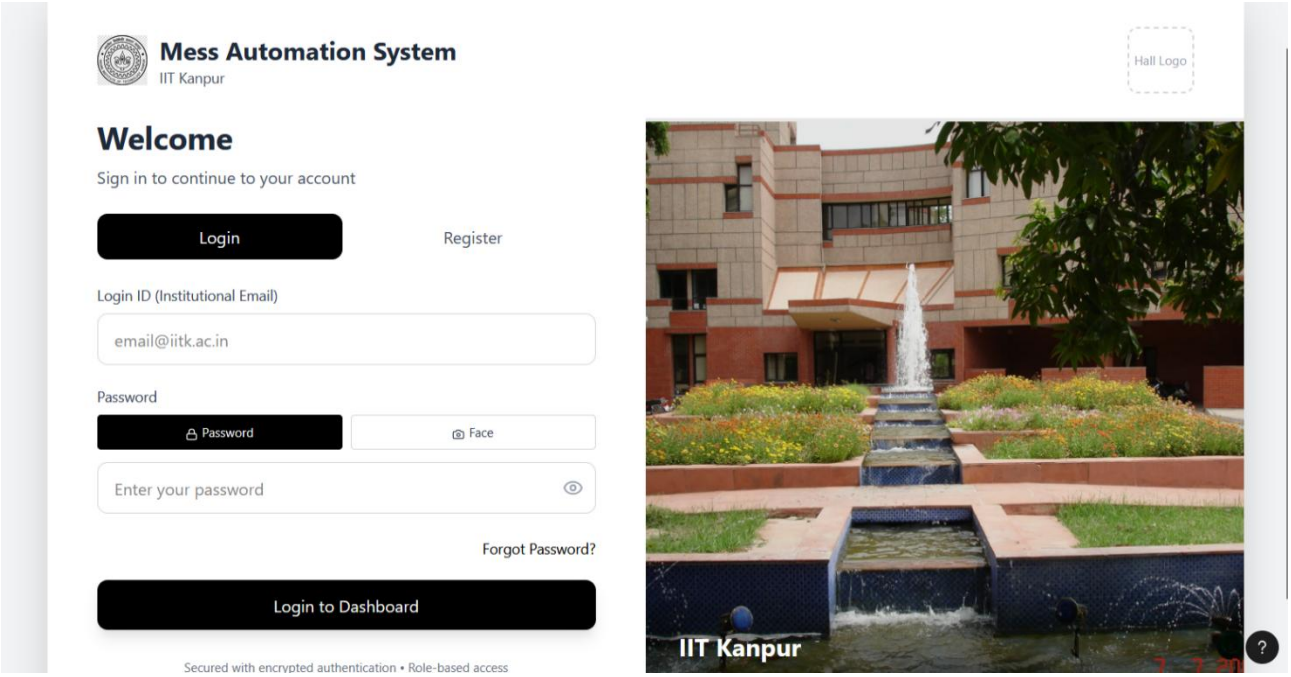
Stable campus network/internet connection.

3 Specific Requirements

3.1 External Interface Requirements

- Login:

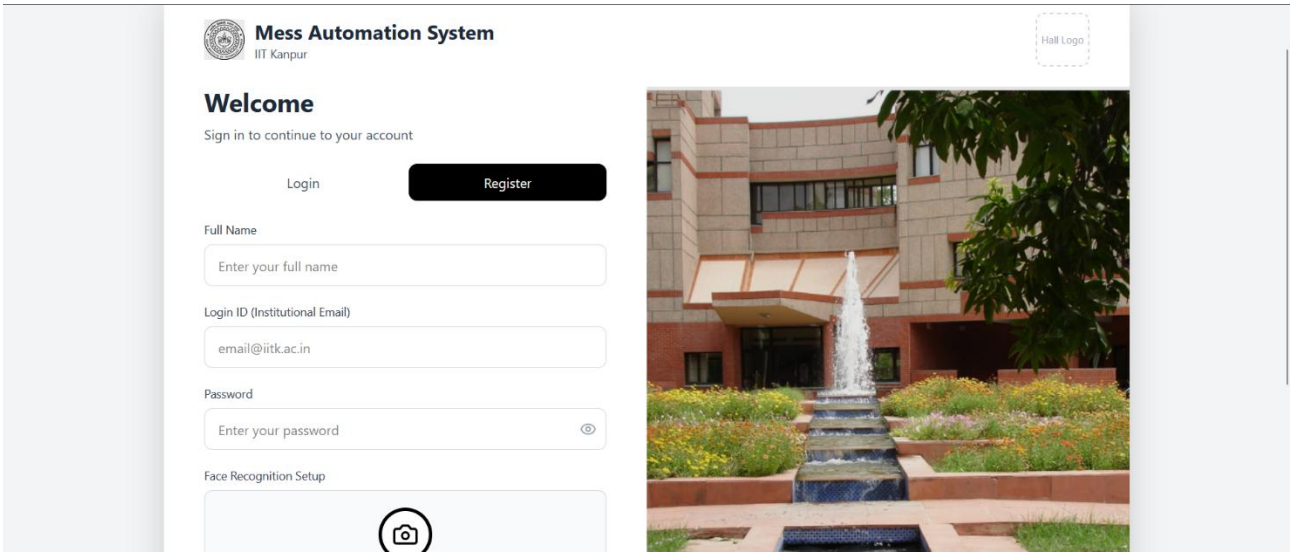
- The login page will ask users to provide their registered **institutional email ID** and **password**.
- Upon successful authentication, users will be redirected to the **home page/dashboard** based on their role (student or mess manager).
- The system supports **optional face-recognition–based authentication** for faster and more secure access.
- Unauthorized or invalid login attempts will display appropriate **error messages**.
- User credentials will be **securely verified** using encrypted authentication mechanisms.
- Role-based access control ensures that **students and mess managers** are directed to their respective dashboards.



1.1.Sign Up

- Allows new students to register and access the Mess Automation System.

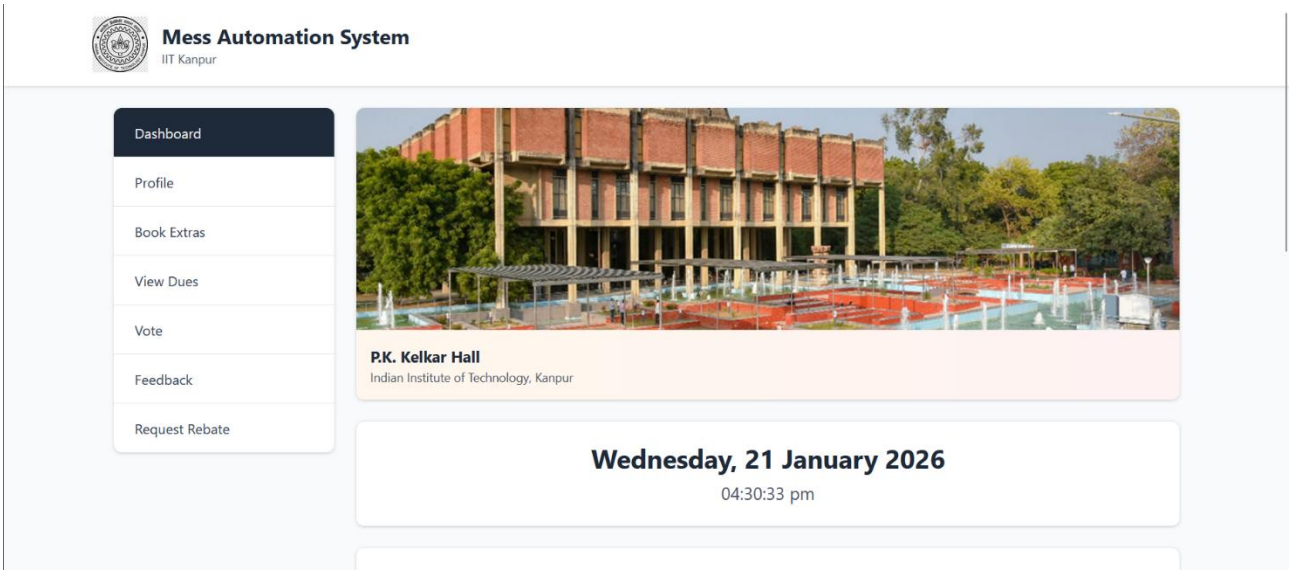
- Students register using **institutional credentials** for secure access.
- Required details: name, roll number, email, and password.
- Enables **face-recognition enrollment** for faster and secure login (optional).
- Once registered, students can:
    - Access mess menus.
    - Purchase extra food items.
    - Vote for monthly menus.
    - Provide feedback.
- Ensures **secure storage of student information** and personalized dashboards.



1.1.Forgot Password

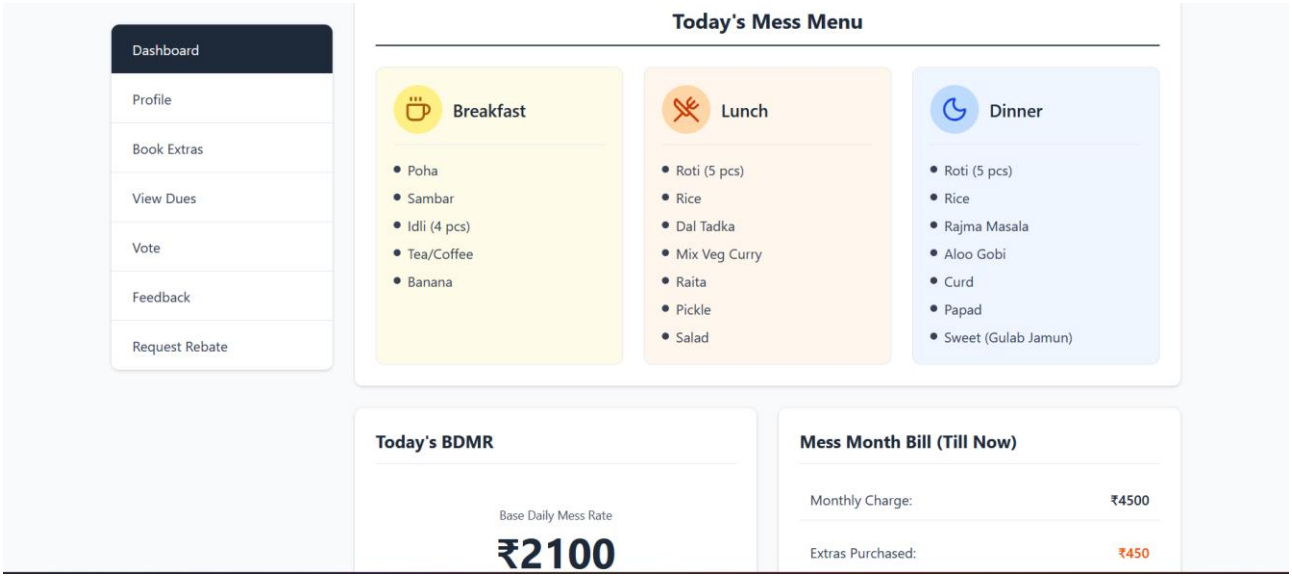
- Helps students recover their account if login credentials are forgotten.
- Recovery is done using the **registered email or institutional ID**.
- Sends a **secure link or verification code** for password reset.
- Ensures students can regain access **without administrative intervention**.
- Maintains **security and privacy** of student accounts.

STUDENT VIEW



1. Home Page (Dashboard)

The **Home Page** serves as the central dashboard of the *Mess Automation System*. It provides users with a quick overview of their mess-related information, including the daily menu, billing status, and navigation options for other features.



1.1 Navigation Sidebar

The left-hand sidebar allows users to easily navigate through different modules of the system:

- **Dashboard** – Displays the home page overview.
- **Profile** – View and update personal details.
- **Book Extras** – Purchase additional food items.
- **View Dues** – Check outstanding payments.
- **Vote** – Participate in mess-related polls.
- **Feedback** – Submit feedback regarding food or services.
- **Request Rebate** – Apply for mess rebates.

This sidebar ensures quick access to all major functionalities.

1.2 Hall Information Section

At the top of the dashboard, the system displays the hall the student belongs to and the institute name.An image of the hall is shown to provide visual identification and improve user experience.

1.3 Date and Time Display

The current **date and time** are prominently displayed below the hall information.
This ensures users are aware of the real-time status when viewing menus or billing details.

1.4 Today's Mess Menu

This section presents the **daily menu**, categorized into three meals:Breakfast , Lunch and Dinner.

Each meal is displayed in a separate card for clarity and ease of reading.

1.5 Today's BDMR

This card shows the **Base Daily Mess Rate (BDMR)** for the current day.
It helps students understand the fixed daily mess cost.

1.6 Mess Month Bill (Till Now)

This section provides a summary of the user's **monthly mess expenses**, including:

a.Mess Monthly Bill

b.Extras purchased

c.Total Dues

This real-time bill breakdown improves transparency and helps users track their spending.

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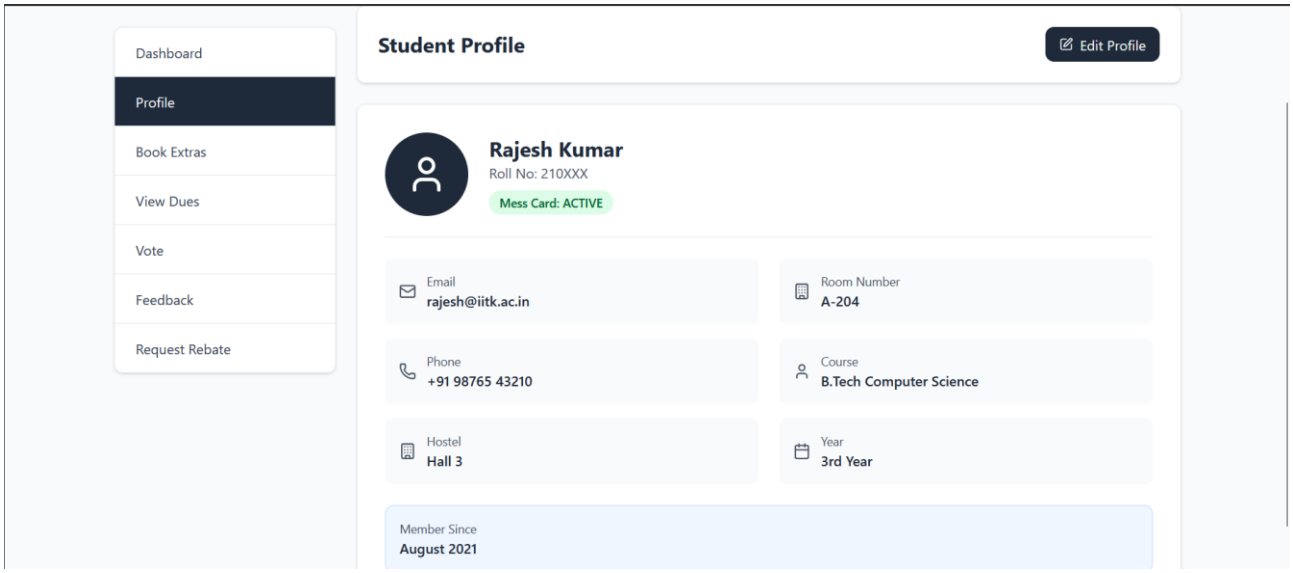
2. *Student Profile Page*

The **Student Profile Page** displays the personal and academic details of a student registered in the *Mess Automation System*. It allows students to view their information and update it when required.



2.1 Profile Header

At the top of the page:

- The title "**Student Profile"** clearly indicates the purpose of the page.
- An **Edit Profile** button is provided, allowing students to modify editable details such as contact information.

2.2 Student Identity Section

This section displays the student's basic identity information like Name , Roll Number and Mess Card Status. A profile icon is shown for visual representation. The mess card status helps confirm whether the student is currently eligible to use mess services.

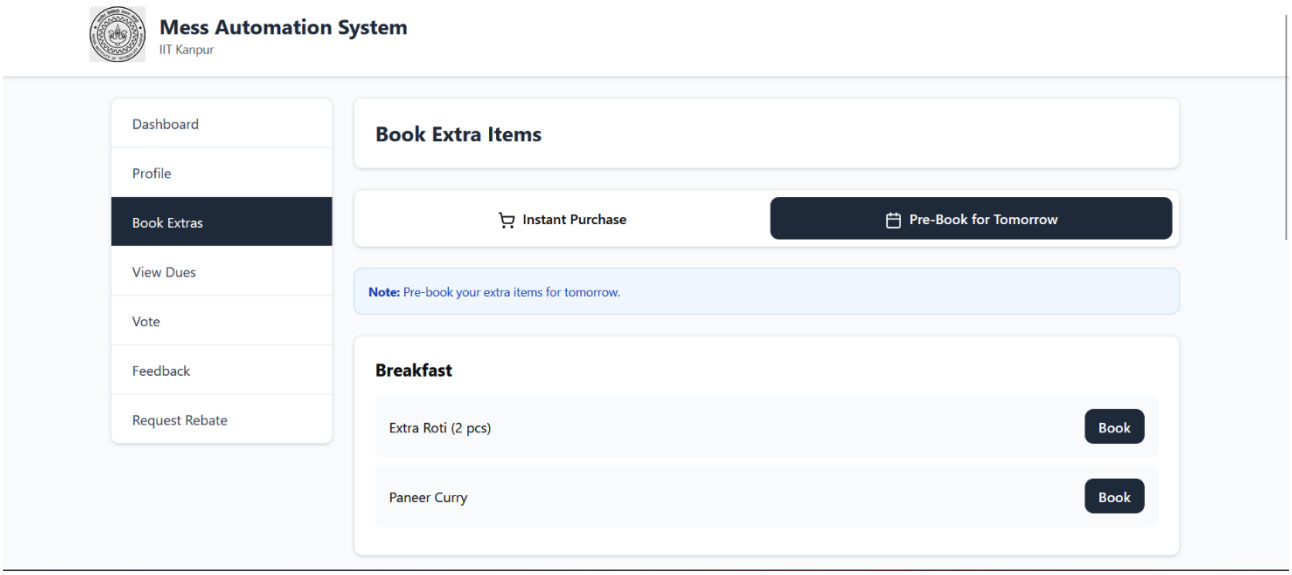
2.3 Personal and Academic Details

The profile details are neatly arranged in card format for better readability:This section provides a complete overview of the student's academic and residential information.

2.4 Membership Information At the bottom of the profile, the system displays membership information.This indicates the duration of the student's association with the mess system.

3. Book Extra Item Page

The **Book Extra Items** page allows students to purchase additional food items apart from the regular mess menu. This feature provides flexibility to students who wish to order extra portions or special items.



3.1 Purchase Mode Selection

At the top of the page, students can choose the mode of purchase:

- **Instant Purchase** – Items are added and billed immediately.
- **Pre-book** – Items can be ordered in advance for a future meal.

This helps students plan their purchases conveniently.

3.2 Extra Items List

Extra food items are displayed in categorized sections for easy browsing.

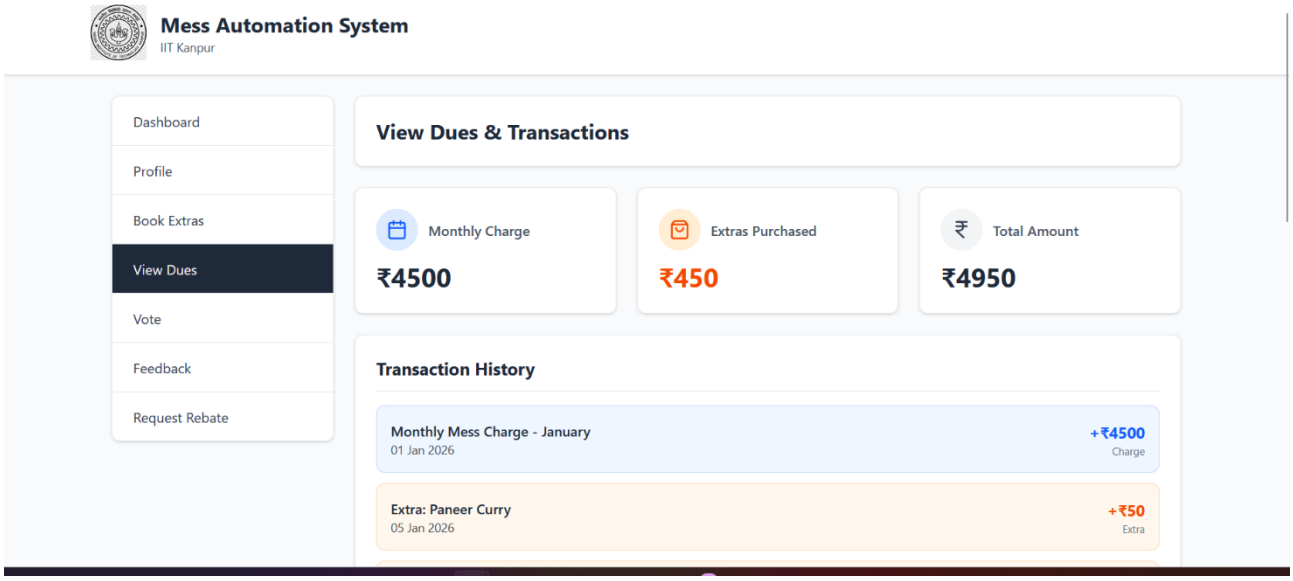
Each item shows its price and includes an **Add** button, allowing students to add the item to their cart.

3.3 Cart Section

The **Your Cart** panel on the right side displays the items selected by the student. Initially, the cart shows that it is empty. As items are added, the cart updates dynamically to reflect selected items and their total cost.

4.View Dues & Transactions Page

The **View Dues & Transactions** page provides students with a detailed overview of their mess-related financial records. It displays the current dues, extra purchases, and a complete transaction history in a transparent and organized manner.



4.1 Dues Summary Section

At the top of the page, a summary of the student's mess charges is displayed using separate cards:

- **Monthly Charge:** Represents the fixed mess charge for the current month.
- **Extras Purchased:** Indicates the total cost of additional food items purchased during the month.
- **Total Amount:** Shows the total payable amount, calculated as the sum of the monthly charge and extras.

This section gives a quick financial snapshot at a glance.

4.2 Transaction History

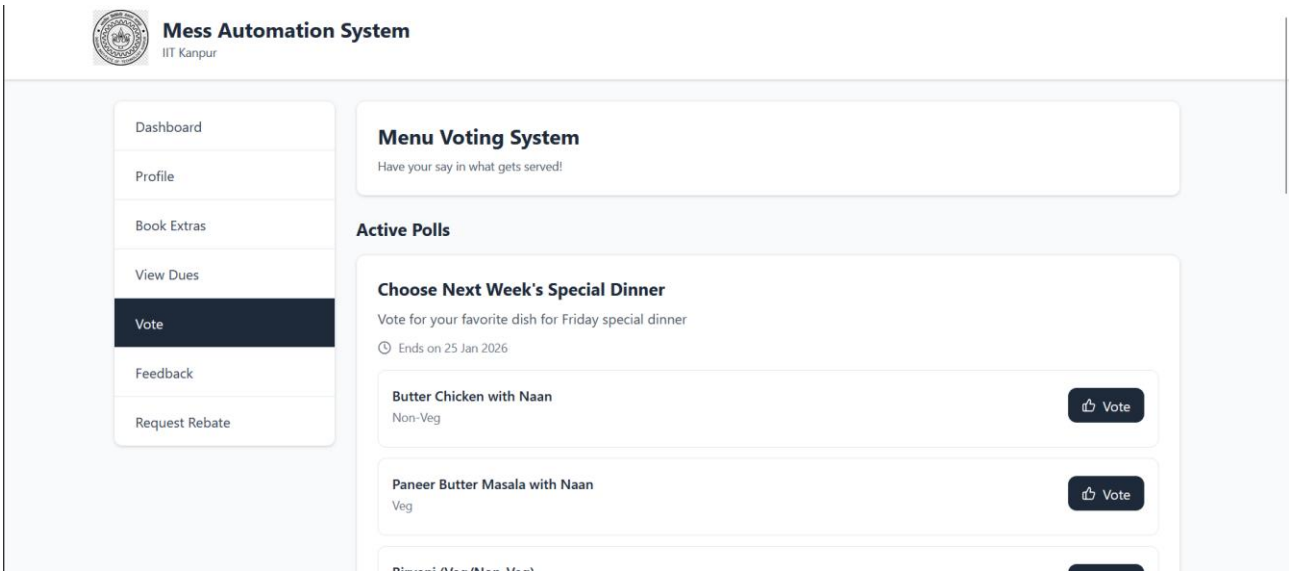
The **Transaction History** section lists all mess-related transactions including charges, extra purchases, and payments.

Each transaction entry includes:

- **Transaction description** (e.g., Monthly Mess Charge, Extra Item purchased, Payment Received)
- **Date of transaction**
- **Amount** with clear indication of charge or payment

5. Menu Voting System Page

The **Menu Voting System** page allows students to participate in decision-making by voting for preferred menu items. This feature improves student involvement and helps the mess management plan meals based on majority preferences.



5.1 Active Polls

The **Active Polls** section displays ongoing polls where students can currently cast their votes.

Example:

- **Poll Title:** Choose Next Week's Special Dinner
  Students can choose one of the options they prefer to have for dinner.

- Each option has a **Vote** button, allowing students to submit their choice. Once submitted, the vote is recorded and cannot be changed.

5.2 Closed Polls

The **Closed Polls** section shows polls that have already ended and whose results are finalized.

Example:

- **Poll:** Sunday Breakfast Menu
- **Winning Item:** Paratha with Curd

This section also displays:

- Total votes received by each option
- Percentage share of votes

This ensures transparency in how menu decisions were made.

5.3 Upcoming Polls

The **Upcoming Polls** section lists polls that are scheduled to open in the future.

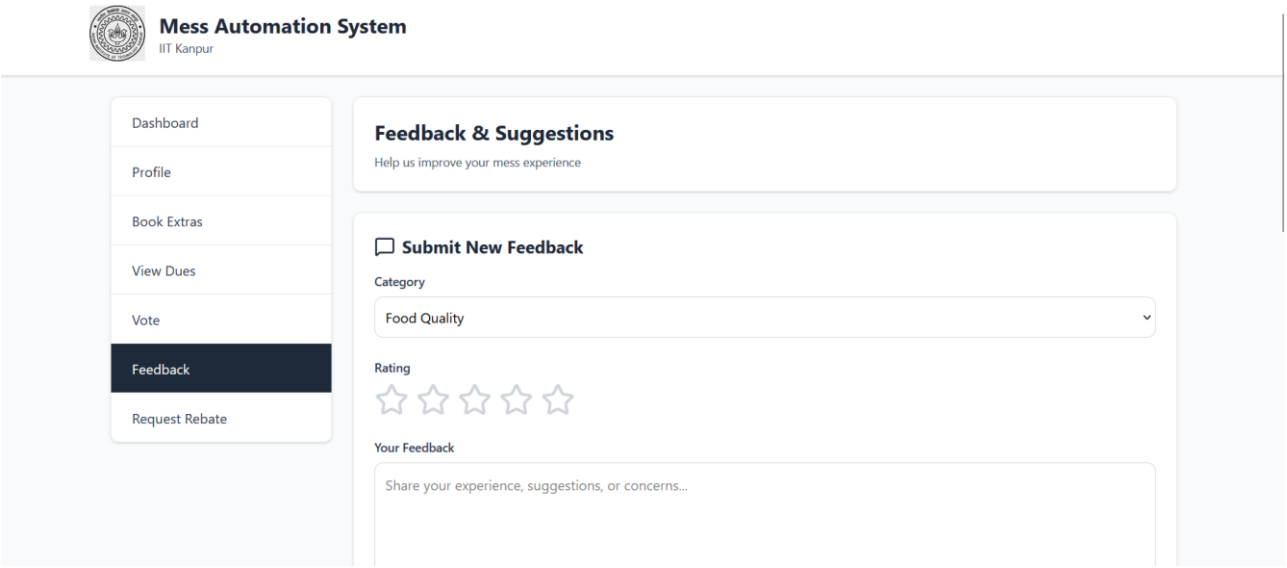
Example:

- **Mid-Semester Exam Week Special**

Students can view upcoming options but cannot vote until the poll becomes active.

6. Feedback & Suggestions Page

The **Feedback & Suggestions** page allows students to share their opinions, experiences, and suggestions regarding mess services. This feature helps mess management identify issues, improve food quality, and enhance overall service.



6.1 Submit New Feedback

This section enables students to submit new feedback through a structured form:

- **Category Selection:**
  Students can choose the type of feedback, such as:
  - Food Quality
  - Service
  - Cleanliness
- **Rating System:**
  A star-based rating system allows students to rate their experience.
- **Feedback Text Box:**
  Students can enter comments, suggestions, or complaints describing their experience.
- **Submit Button:**
  The **Submit Feedback** button records the feedback and stores it in the system.

This structured approach ensures clear and actionable feedback.

6.2 Previous Feedback

The **Your Previous Feedback** section displays feedback previously submitted by the student.

Each entry includes:

- Feedback category
- Date of submission
- Star rating
- Feedback message

This allows students to track their submissions and ensures transparency.

6.3 Feedback Guidelines

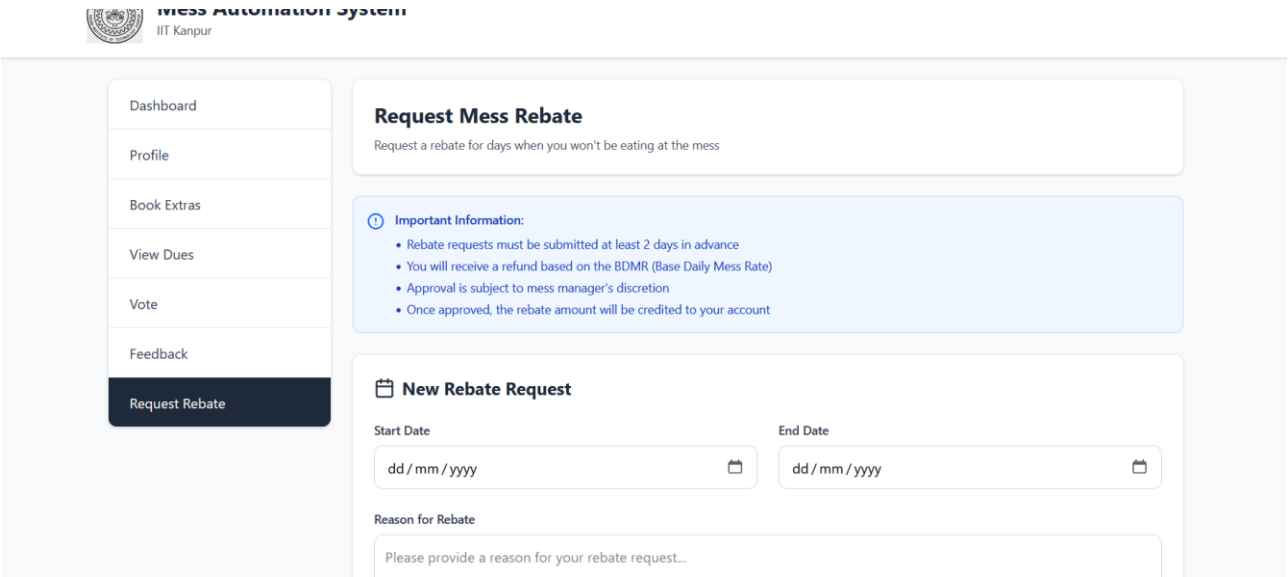
At the bottom of the page, guidelines are displayed to help students submit meaningful feedback, such as:

- Be clear and specific in describing issues
- Avoid offensive language
- Submit feedback responsibly

These guidelines help maintain constructive communication.

7. Request Mess Rebate Page

The **Request Mess Rebate** page allows students to apply for a rebate when they are not availing mess services for a certain period. This feature ensures fair billing by adjusting charges for approved absence durations.



7.1 Important Information Section

At the top of the page, important instructions related to rebate requests are displayed, such as:

- Rebate requests must be submitted in advance
- A minimum number of consecutive days is required for eligibility
- Approved rebates are adjusted in the Base Daily Mess Rate (BDMR)

This helps students understand the rules before submitting a request.

7.2 New Rebate Request Form

Students can submit a new rebate request using the following fields:

- **Start Date:** Beginning date of absence
- **End Date:** Ending date of absence
- **Reason for Rebate:** Text box to explain the reason for not availing mess services

After filling in the details, students can submit the request using the **Submit Rebate Request** button.

7.3 Previous Rebate Requests

The **Previous Rebate Requests** section displays the history of rebate requests submitted by the student.

Each entry includes:

- Rebate period (start and end dates)
- Status of the request (e.g., Approved)

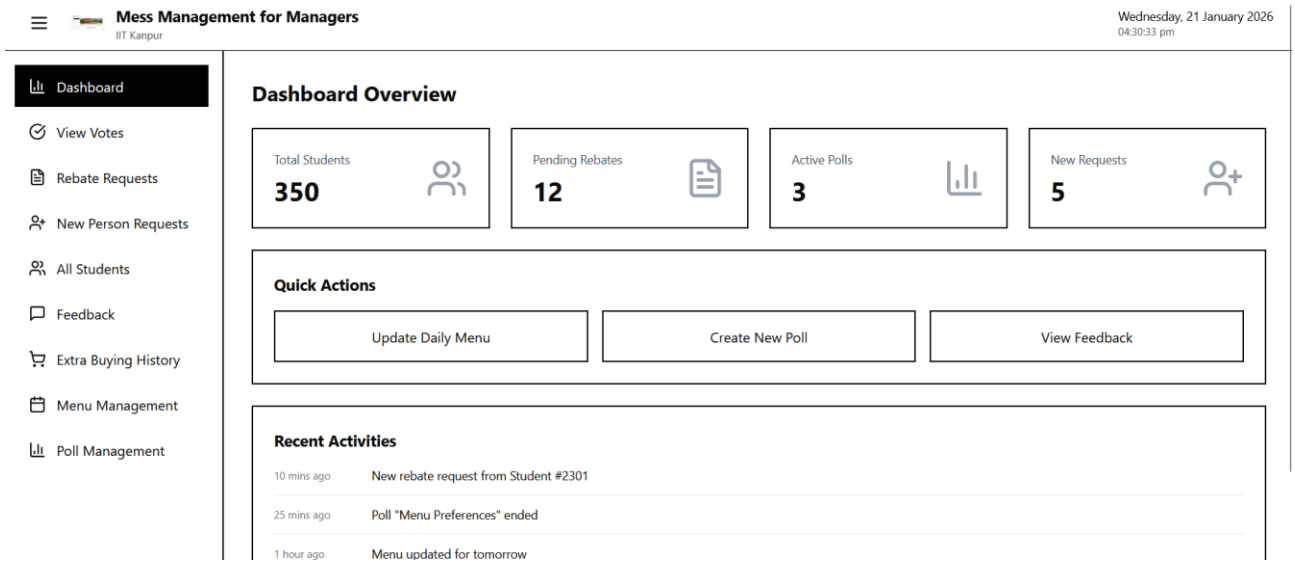
This allows students to track the approval status of their rebate applications.

MANAGER VIEW

1. Manager Dashboard

The **Manager Dashboard** is the main overview page for mess administrators. It provides a snapshot of current mess operations and quick access to important actions.

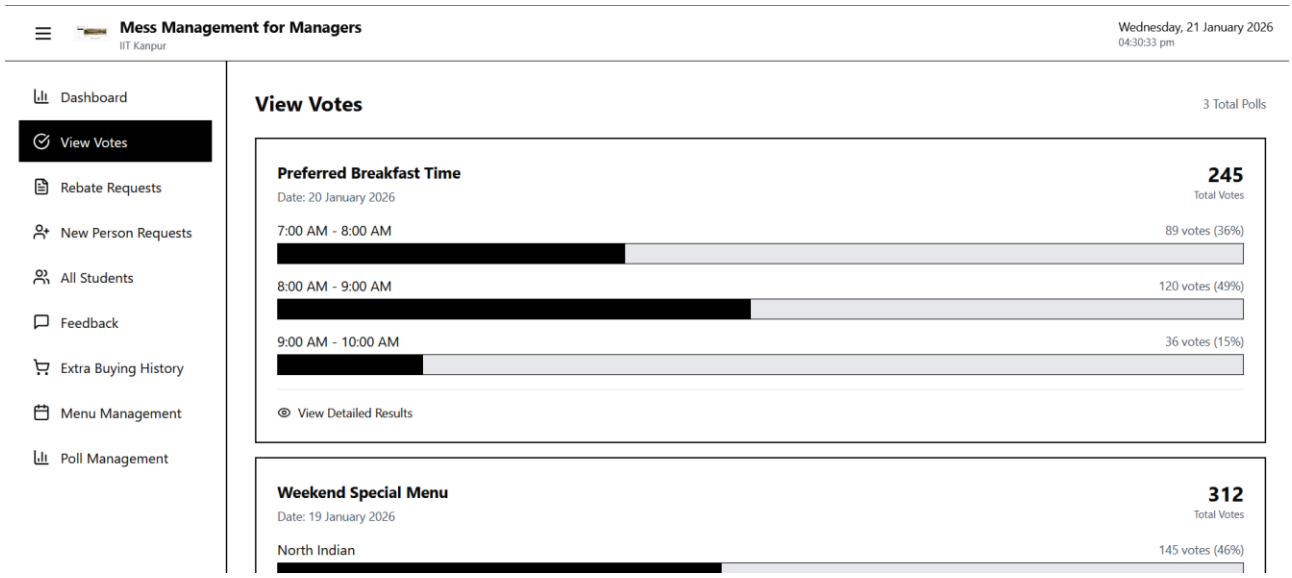
- **Summary Cards:** Show key statistics such as total students, pending rebate requests, active polls, and new requests.
- **Quick Actions:** Allow managers to update the daily menu, create new polls, and view student feedback easily.
- **Recent Activities:** Displays a timeline of recent events like rebate submissions, poll updates, and menu changes.
- **Sidebar Navigation:** Provides access to modules such as votes, rebate requests, student management, feedback, menu management, and poll management.



2.View Votes – Brief Description

The **View Votes** page allows mess managers to see the results of polls conducted among students.

- Displays a list of polls with their **title, date, and total votes**.
- Each poll shows **options with vote counts and percentages** using progress bars for easy comparison.
- Helps identify the **most preferred choices** (e.g., breakfast time slots, menu preferences).
- Includes an option to **view detailed results** for deeper analysis.
- Sidebar navigation allows quick access to other management features.

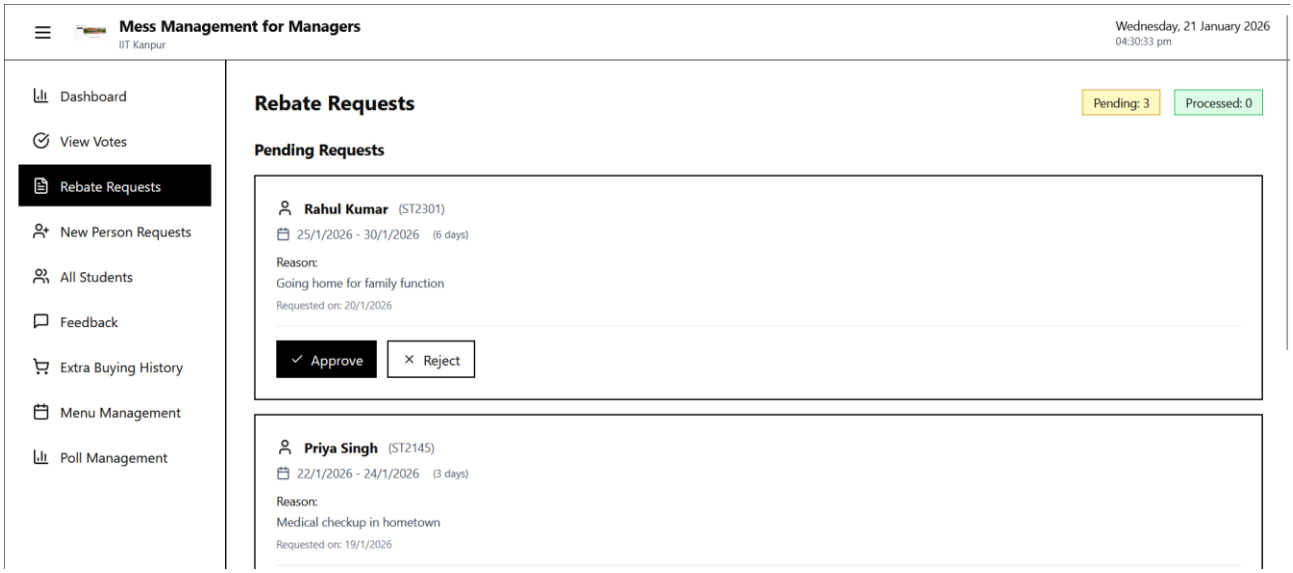


3. Rebate Requests

The **Rebate Requests** page allows mess managers to review and process student rebate applications.

- Displays the number of **pending** and **processed** requests.
- Lists **pending rebate requests** with student details, rebate duration, reason, and request date.
- Provides **Approve** and **Reject** actions for each request.
- Ensures transparent and efficient handling of mess rebates.

This page helps managers quickly verify requests and take timely action.

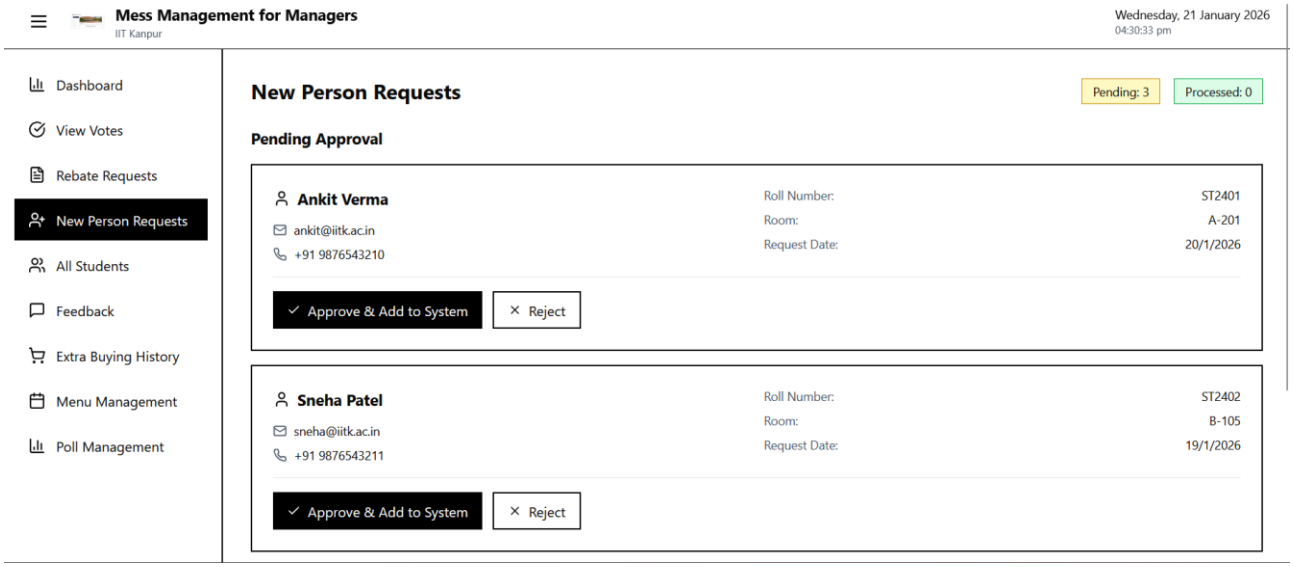


4. New Person Requests – Brief Description

The **New Person Requests** page allows mess managers to review and approve new student registrations.

- Displays counts of **pending** and **processed** requests.
- Lists pending requests with **student details** such as name, roll number, room, email, phone number, and request date.
- Provides actions to **Approve & Add to System** or **Reject** each request.
- Ensures only verified students are added to the mess system.

This page helps managers control and validate new entries efficiently.

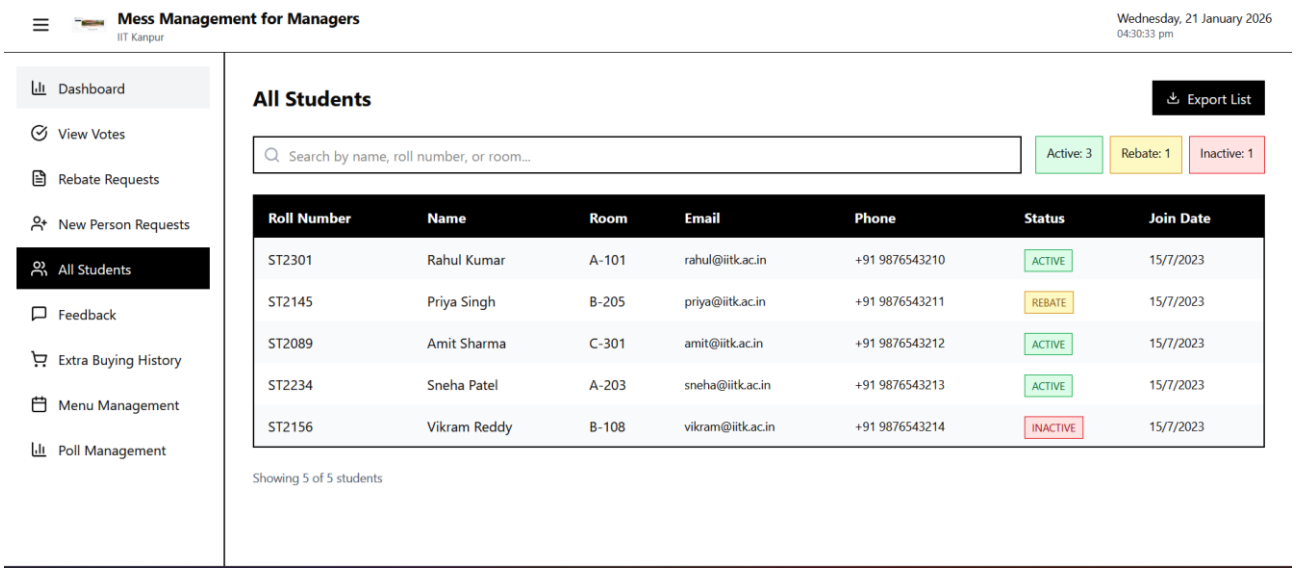


5.All Students – Brief Description

The **All Students** page provides a complete list of students registered in the mess system.

- Displays student details including **roll number, name, room, email, phone, status, and join date**.
- Includes a **search bar** to find students by name, roll number, or room.
- Shows student **status indicators** such as Active, Rebate, and Inactive.
- Provides quick **status counts** and an option to **export the student list**.

This page helps managers efficiently track and manage all registered students.

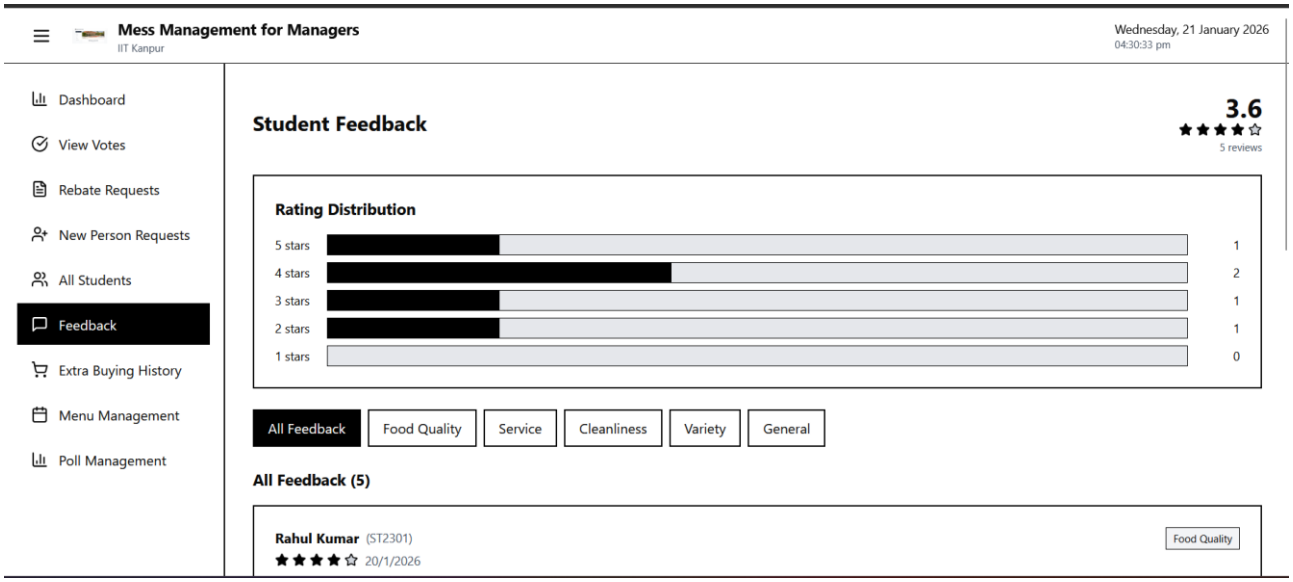


6. Student Feedback

The **Student Feedback** page displays ratings and reviews submitted by students about the mess.

- Shows the **overall average rating** and total number of reviews.
- Displays a **rating distribution** (1–5 stars) using progress bars.
- Allows filtering feedback by categories such as **Food Quality, Service, Cleanliness, Variety, and General**.
- Lists individual feedback entries with **student details, rating, date, and category**.

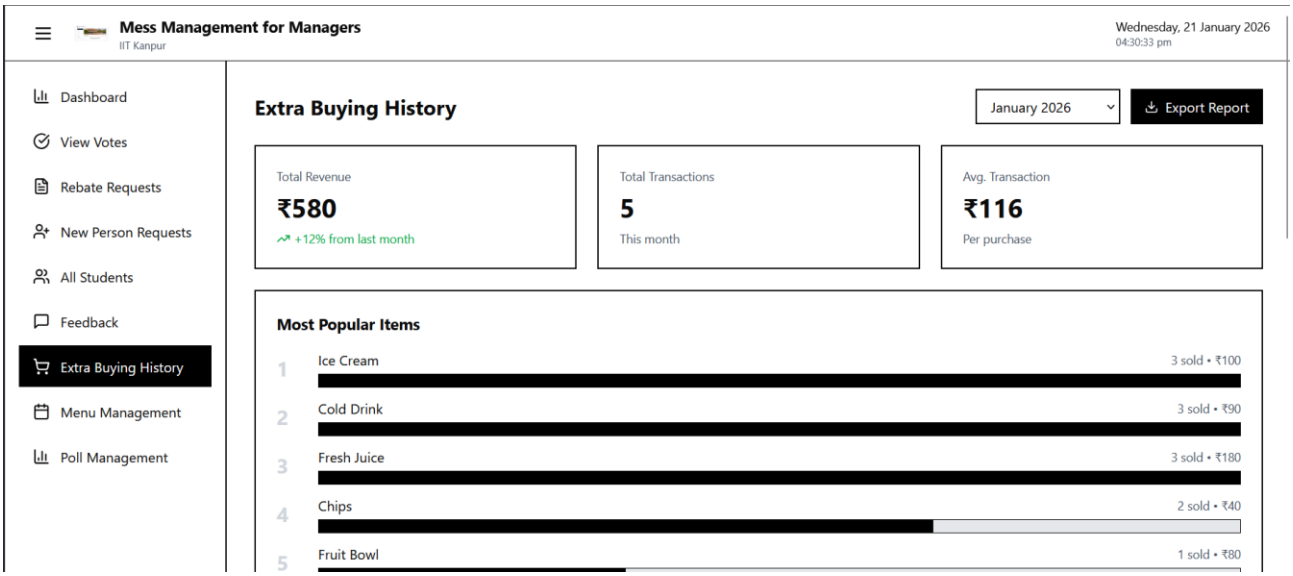
This page helps managers understand student satisfaction and identify areas for improvement.



7. Extra Buying History

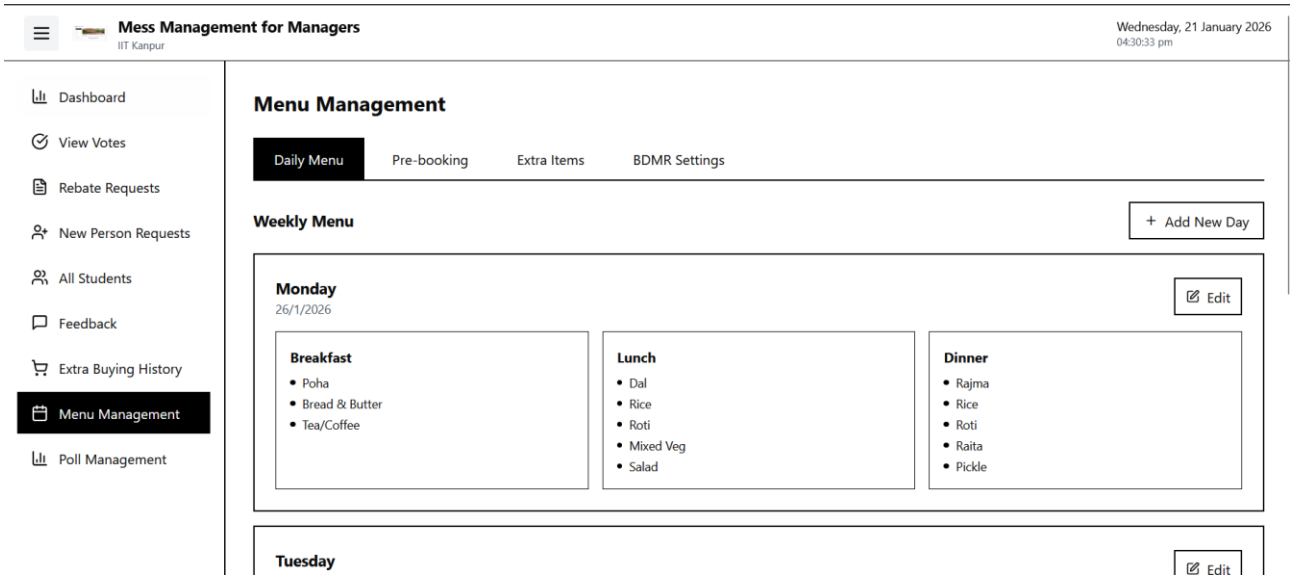
The **Extra Buying History** page provides a complete overview of additional food purchases made by students.

- Displays key metrics such as **total revenue**, **number of transactions**, and **average transaction value** for a selected month, with an option to **export reports**.
- Shows **most popular extra items** along with quantities sold and revenue generated.
- Lists **recent purchases** with student details, purchase date, items bought, quantities, and total amount spent.



8.Menu Management Page – Brief

- Used to create and update mess menus
- Accessible only to mess managers
- Menu is organized week-wise and day-wise
- Each day includes Breakfast, Lunch, and Dinner items
- Tabs available for Daily Menu, Pre-booking, Extra Items, and BDMR Settings
- Edit option allows updating menu for a specific day
- Add New Day option allows extending the weekly menu
- Ensures structured and efficient menu planning

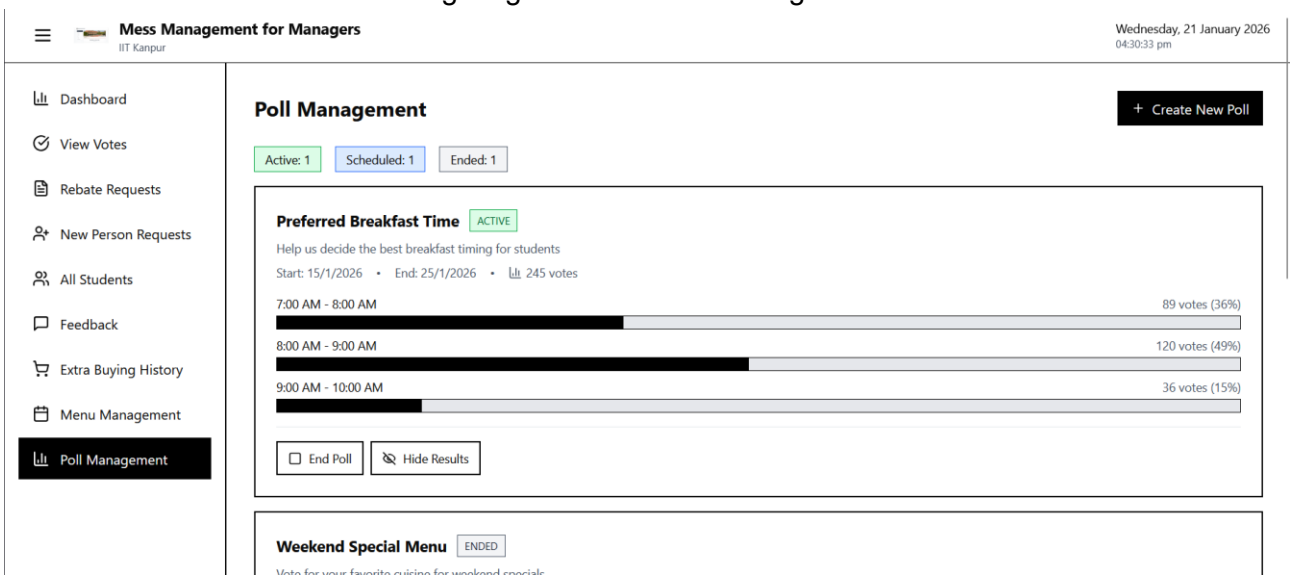


9. Poll Management Page

This page helps managers analyze sales trends, monitor transactions, and plan inventory effectively.

- Displays the **Poll Management** section for mess managers
- Shows count of **Active, Scheduled, and Ended** polls
- Lists polls with **title, description, status, start/end dates, and total votes**
- Visualizes voting results using **progress bars and percentages**
- Allows managers to **end polls** or **hide results**
- Provides a **Create New Poll** button for adding new polls

Includes a **sidebar menu** for navigating to other mess management features



3.2 Functional Requirements

FR1: User Authentication

- The system shall allow users to log in using registered institutional email ID and password.
- The system shall support optional face-recognition–based authentication.
- The system shall validate user credentials and prevent unauthorized access.
- The system shall provide appropriate error messages for invalid login attempts.

FR2: Role-Based Access Control

- The system shall distinguish between student and mess manager roles.
- The system shall redirect users to role-specific dashboards after login.

FR3: Student Dashboard

- The system shall display the daily menu to students.
- The system shall show extra food purchase history and pending dues.
- The system shall allow students to navigate to extras purchase, voting, and feedback modules.

FR4: Extra Food Purchase Management

- The system shall allow students to view available extra food items with prices.
- The system shall allow students to purchase extra items.
- The system shall record all extra food transactions in the database.
- The system shall update the student's dues accordingly.

FR5: Menu Voting System

- The system shall allow students to vote for preferred dishes during the voting period.
- The system shall display only feasible dish options.
- The system shall restrict students to one vote per voting cycle.
- The system shall store voting results securely.

FR6: Mess Manager Controls

- The system shall allow mess managers to add, update, or remove menu items.
- The system shall allow mess managers to manage raw material inventory.
- The system shall allow mess managers to update prices of extra food items.
- The system shall allow mess managers to open and close voting windows.

FR7: Feedback Management

- The system shall allow students to submit feedback on food quality and services.
- The system shall store feedback securely.
- The system shall allow mess managers to view submitted feedback.

FR8: Automated Reports

- The system shall generate monthly reports of extra food purchases.

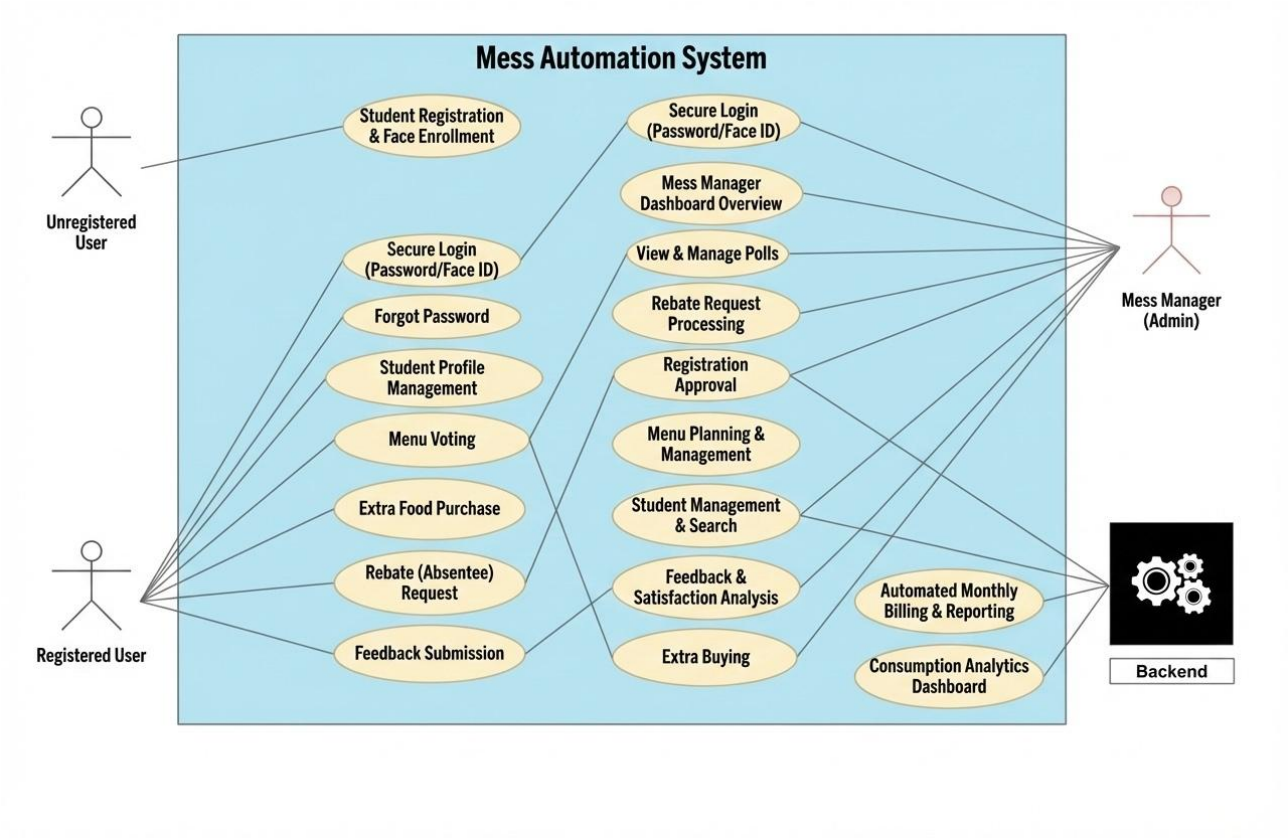
FR9: Data Management

- The system shall store user data, transactions, votes, and feedback in a centralized database.
- The system shall ensure data consistency and integrity.

FR10: Analytics Support

- The system shall generate basic consumption analytics for mess managers.
- The system shall display summarized usage statistics on the admin dashboard.

3.3 Use Case Model



3.3.1 Student Registration & Face Enrollment

Purpose	To register new students and enroll their biometric data for mess entry.
Requirements Traceability	Name, Roll No, IITK Email ID, Biometric Face ID data.

Priority	High
Preconditions	Students must have a valid institute email ID (@iitk.ac.in).
Post conditions	The student is registered, and Face ID data is stored for authentication.
Actors	Student, System
Exceptions	1. If the email is not in the @iitk.ac.in domain, the system rejects registration. 2. If Face ID enrollment fails (e.g., poor lighting), the system prompts the student to retry.
Includes	None
Notes/Issues	1. Biometric data must be encrypted for privacy and stored securely. 2. System must ensure that a single Roll No cannot be registered multiple times.

3.3.2 Secure Login (Password/Face ID)

Purpose	To authenticate users (Students/Admin) into the system using multi-modal methods.
Requirements Traceability	Registered Roll No/Email and Password, Biometric Face ID data, User Role (Student/Admin).

Priority	High
Preconditions	User must be registered in the system.
Post conditions	User is granted access and redirected to their specific dashboard.
Actors	Student, Mess Manager (Admin), System
Exceptions	1. Face not recognized: System prompts for manual password entry. 2. Invalid credentials: System displays "Incorrect Credentials" and denies access. 3. After three failed attempts, the system locks the account for 10 minutes.
Includes	None
Notes/Issues	Ensure Face ID works in low light or suggest password fallback.

3.3.3 Forgot Password

Purpose	To allow students to securely reset their login credentials without administrative help.
Requirements Traceability	Registered Institutional Email ID, Verification Code (OTP) or Secure Reset Link.
Priority	Medium

Preconditions	The student must have a previously registered and verified account in the system.
Post conditions	The student's old password is invalidated, and the new password is encrypted and stored.
Actors	Student
Exceptions	1. Invalid Email: If the entered email is not in the database, the system displays "User not found." 2. Expired Link: If the student clicks the reset link after the time limit, the system prompts them to request a new one.
Includes	None
Notes/Issues	Password must meet complexity requirements (e.g., minimum 8 characters).

3.3.4 Student Profile Management

Purpose	To allow students to view their personal, academic, and membership details and update editable information.
Priority	Medium
Requirements Traceability	Password
Preconditions	Student must be logged into the system.

Post conditions	Student's profile is updated in the database and changes are reflected in the UI.
Actors	Student
Exceptions	If the student attempts to edit read-only fields (like Roll Number or Joining Date), the system prevents the action.
Includes	None

3.3.5 Menu Voting

Purpose	To allow students to participate in deciding the menu via a democratic voting system.
Requirements Traceability	None
Priority	High
Preconditions	Admin has enabled the voting window for the upcoming week.
Post conditions	Votes are tallied, and the final menu is generated based on the results.
Actors	Student, Mess Manager (Admin), System

Exceptions	1. Voting deadline passed: Student receives a "Voting Closed" notification. 2. Duplicate voting: Students are prevented from voting more than once for the same meal.
Includes	Use Case #2 (Login)
Notes/Issues	Admin should have the power to override votes in case of logistics/supply issues.

3.3.6 Extra Food Purchase

Purpose	To allow students to buy items not included in the basic menu (e.g., eggs, milk, special dishes).
Requirements Traceability	None
Priority	High
Preconditions	Student is logged in; Items must be listed in the current "Extras" menu.
Post conditions	The student's digital ledger/bill is updated with the purchase amount.
Actors	Student, System

Exceptions	Out of Stock: System prevents selection of unavailable items.
Includes	None
Notes/Issues	Real-time update of inventory is crucial to avoid selling out-of-stock items.

3.3.7 Rebate(Absentee) Request

Purpose	To allow students to mark themselves as absent for specific meals to save food and reduce their monthly bill.
Requirements Traceability	Start Date and End Date of absence, Reason for leave (Optional).
Priority	High
Preconditions	Student must be logged into the system. The request must be made at least 24 hours (or as per mess policy) before the meal.
Post conditions	The student is marked 'Absent' for the selected dates. The system updates the expected headcount for the cook and calculates the rebate credit for the bill.
Actors	Student, System

Exceptions	1. Late Submission: If the student tries to apply for a rebate after the deadline, the system shows "Deadline Expired." 2. Duration Limit: If the rebate exceeds the maximum allowable days per month, the system notifies the student to contact the Admin.
Includes	None
Notes/Issues	The system must provide a summary of total rebate days to the Mess Manager daily for procurement adjustments.

3.3.8 Feedback Submission

Purpose	To provide a channel for students to rate meal quality and for Admins to address grievances.
Requirements Traceability	None
Priority	Medium
Preconditions	Student must have consumed the meal (authenticated at entry).
Post conditions	Feedback is stored and categorized; Admin is notified of low-rated meals.
Actors	Student, System, Mess Manager (Admin)

Exceptions	Duplicate Feedback: Students are prevented from rating the same meal multiple times.
Includes	None
Notes/Issues	Sentiment analysis could be added later to categorize comments automatically.

3.3.9 Automated Monthly Billing & Reporting

Purpose	To calculate the final mess bill for each student, accounting for base fees, extras, and rebates.
Requirements Traceability	None
Priority	High
Preconditions	End of the billing cycle (typically the last day of the month).
Post conditions	Students are notified via the app/email.
Actors	System, Mess Manager (Admin)
Exceptions	Calculation Error: If data for a specific day is missing, the system flags the student record for manual Admin review.
Includes	None

Notes/Issues	The formula for "Rebate Credit" must be configurable by the Admin.

3.3.10 Consumption Analytics Dashboard

Purpose	To provide the Mess Manager with data-driven insights into food wastage and popularity.
Requirements Traceability	None
Priority	Medium
Preconditions	Admin is logged into the system.
Post conditions	Graphs and charts representing mess data are displayed.
Actors	Mess Manager (Admin), System
Exceptions	Insufficient Data: If no votes or attendance records exist for the period, the system displays a "No Data Available" message.
Includes	None
Notes/Issues	This dashboard is critical for the "Automation" part of the project to justify cost-saving.

3.3.11 Mess Manager Dashboard Overview

Purpose	To provide a snapshot of mess operations and quick access to administrative modules.
Requirements Traceability	None
Priority	High.
Preconditions	Manager must be logged into the system.
Post conditions	Manager gains situational awareness of current mess status and can navigate to specific management tasks.
Actors	Mess Manager, System.
Exceptions	If database connection fails, Summary Cards display "N/A" or error messages.
Includes	None

3.3.12 View & Manage Polls

Purpose	To conduct student polls, monitor voting trends, and analyze results for menu planning.
Requirements Traceability	Password

Priority	High.
Preconditions	Manager is logged in and navigated to Poll Management or View Votes page.
Post conditions	A new poll is created or existing poll results are used to inform mess decisions.
Actors	Mess Manager.
Exceptions	If no votes are recorded, system displays "0 votes" and empty progress bars.
Includes	None

3.3.13 Rebate Request Processing

Purpose	To review and systematically process student rebate applications.
Requirements Traceability	Student details (Name, Roll No, Room, Email, Phone).
Priority	High.
Preconditions	Students have submitted rebate requests through the system.
Post conditions	Rebate status is updated, and the student's bill and mess headcount are adjusted accordingly.

Actors	Mess Manager.
Exceptions	If a request is missing critical data, manager may reject it with a note.
Includes	None

3.3.14 Registration Approval

Purpose	To verify and validate new student entries into the mess system.
Requirements Traceability	Student details (Name, Roll No, Room, Email, Phone).
Priority	High.
Preconditions	New students have submitted a registration request.
Post conditions	Verified student is added to the active database and can access the app.
Actors	Mess Manager.
Exceptions	If Roll Number is invalid, manager rejects the request.
Includes	None

3.3.15 Menu Planning & Management

Purpose	To create, edit, and organize weekly mess menus.
Requirements Traceability	None
Priority	High.
Preconditions	Manager is logged in.
Post conditions	The updated menu is visible to all registered students.
Actors	Mess Manager.
Exceptions	If mandatory meal slots are left empty, system prompts for input before saving.
Includes	None

3.3.16 Student Management & Search

Purpose	To track and manage the database of all registered students.
Requirements Traceability	Students' information
Priority	Medium.

Preconditions	Students are already registered in the system.
Post conditions	Manager obtains student lists or exports them for administrative reporting.
Actors	Mess Manager.
Exceptions	If no results match the search, the system displays "No student found".
Includes	None

3.3.17 Feedback & Satisfaction Analysis

Purpose	To monitor student satisfaction and identify mess service improvements.
Requirements Traceability	None
Priority	Medium.
Preconditions	Students have submitted ratings and reviews.
Post conditions	Manager gains insight into performance trends across different mess categories.
Actors	Mess Manager.

Exceptions	None.
Includes	None

3.3.18 Extra Buying

Purpose	To analyze sales trends and monitor revenue from extra food purchases.
Requirements Traceability	Student details (Name, Roll No, Room, Email, Phone).
Priority	High.
Preconditions	Students have made "Extra" purchases through the app.
Post conditions	Monthly financial reports are generated for inventory and sales planning.
Actors	Mess Manager.
Exceptions	If no transactions occurred in the selected period, metrics display as zero.
Includes	None

Other Non-functional Requirements

A. Performance Requirements

1. Reliability

a. It is expected that the system may occasionally fail so, the system shall recover and become operational as soon as possible.

b. The system must remain fully functional during official mess timings as defined by the hall office.

c. Scheduled maintenance shall not be conducted during peak usage hours, including mess timings and voting periods.

2. Response Time

a. Whenever an admin or student searches for any data or information, the system shall respond within a few seconds under normal operating conditions.

b. The software shall load all user interface components and information within fractions of a second .

c. The system shall ensure a smooth user experience with no unusual or unexpected delays during interaction.

d. Common actions such as login, viewing records, and submitting entries shall be done in a few seconds time.

B. Safety and Security Requirements

1. Security

a. User credentials (Admin and Students) shall be securely stored.

b. There shall be no possibility of credential leakage or unauthorized access to user accounts.

c. Users other than Admins shall not be allowed to modify entries that are designated as admin-only operations.

d. The software shall not disclose any information to unauthorized individuals.

e. Users must log in before accessing any system data or functionality.

2. Authentication

a. The methods of authentication for new users shall be defined and managed by the hall office.

b. Once registered, users shall be allowed to change their passwords after verification through their official IITK email ID.

c. The system may provide an alternative authentication method using CC user ID and password, subject to institutional approval.

d. Password change and reset operations shall follow secure verification procedures.

C. Software Quality Attributes

1. Availability

a. The software and its data shall be accessible to authorized users at all times , except during scheduled maintenance.

b. The system shall be capable of supporting most of the residents of the hall during peak hours.

c. Real-time data shall be made available to users immediately whenever the Admin enables such access (eg. during voting periods).

2. Performance

a. The system shall achieve a maximum uptime, excluding scheduled maintenance.

3. Usability

a. Software must be testable, and users should be able to perform operations without confusion.

b. The user interface shall provide clear feedback for almost all actions, including successful operations, warnings, and errors.

c. The overall user experience shall be smooth, intuitive, and free from unnecessary complexity.

Appendix A – Data Dictionary

Student Class:

Element Name	Description	Attributes	Operations	Relationships
Student	Represents a hostel student who uses the mess services and interacts with the system through a student dashboard.	RollNo Name Email Password RoomNo	login() vote_Menu, view_Dashboard make_Purchase submit_Feedback view_Billing	The Student class has a many-to-one relationship with the Menu class and a one-to-many relationship with the Feedback class.

Admin Class:

Element Name	Description	Attributes	Operations	Relationships
Mess Manager	The mess manager responsible for operations and oversight.	AdminID Name Password Role	login() manage_Menu track_Billing monitor_Participation view_Analytics reply_Feedback	The Admin class has a one-to-many relationship with the Student class and the Menu class.

Menu Class:

Element Name	Description	Attributes	Operations	Relationships
Menu	The scheduled food items for various meal times.	MenuID Date MealType (B/L/D) ItemsList, VoteCount	add_Item remove_Item update_Menu reset_Votes	The Menu class has a many-to-one relationship with the Admin class and receives interactions from the Student class.

Bill & Extra Purchase Class:

Element Name	Description	Attributes	Operations	Relationships
Transaction	Records of mess fees and extra purchases made by students.	TransactionID StudentID Amount Date PurchaseType	record_Purchase generate_Bill update_Balance	The Transaction class has a many-to-one relationship with the Student class.

Voting Class:

Element Name	Description	Attributes	Operations	Relationships
Voting	Represents the process of voting for mess menu items by students..	VoteID StudentID MenuID DishID VoteDate VotingCycle	cast_Vote validate_Vote count_Votes close_Voting	The Voting class has a many-to-one relationship with the Student class and is associated with the Menu class. It is monitored by the Admin class. class.

Feedback Class:

Element Name	Description	Attributes	Operations	Relationships
Feedback	Stores feedback provided by students regarding food quality and mess services.	FeedbackID StudentID FeedbackText Rating Category DateSubmitted	submit_Feedback edit_Feedback view_Feedback respond_Feedback	The Feedback class has a many-to-one relationship with the Student class and is reviewed by the Admin class.

Appendix B - Group Log

Date	Summary of the Meeting
7th January 2026	Discussion on different new project ideas proposed by the group members.
8th January 2026	Detailed discussion of every idea and shortlisting the top three.
9th January 2026	Shortlisted ideas are discussed with the professor and Finalized “Mess Automation System” as our final idea.
15th January 2026	Discussion on the Software Requirement Specification and task allocation by forming pairs.
17th January 2026	Met our project mentor TA Ujjwal Kajal to share our ideas.
18th - 22nd January 2026	Working on SRS documentation
22nd January 2026	Met TA and taken some inputs and modifications .Finalized SRS document